

iClusterBayes Summary Report

Automated Report

Hyperparameters

```
cat('sdev = 0.01, beta.var.scale = 0.2\n')
```

```
## sdev = 0.01, beta.var.scale = 0.2
```

Key Summary Statistics

```
cat(paste0('```\n', '
--- Summary for K = 1 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.6850  0.7078  0.7144  0.7149  0.7222  0.7439

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.09278 0.13722 0.13204 0.16778 0.84556
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.09556 0.14000 0.13478 0.17111 0.76389
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005000 0.008333 0.101966 0.031111 1.000000
```

```

Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0005556 0.0050000 0.0083333 0.0773990 0.0244444 1.0000000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0061111 0.0150000 0.162394 0.118889 1.000000

--- Summary for K = 2 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
 0.4622  0.4967  0.5044  0.5040  0.5117  0.5367

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.04778 0.09722 0.09920 0.13611 0.91000
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.0500  0.1011  0.1020  0.1400  0.6922
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.001667 0.058333 0.413768 1.000000 1.000000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0005556 0.0050000 0.2480509 0.3904167 1.0000000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.1228  1.0000  0.6968  1.0000  1.0000

--- Summary for K = 3 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
 0.4294  0.4550  0.4644  0.4642  0.4728  0.5067

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.02222 0.05944 0.06859 0.09833 0.89111
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.02333 0.06222 0.07092 0.10222 0.60056
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.4672  0.7611  0.6738  1.0000  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.7706  1.0000  1.0000  0.9998  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0000000 0.0005556 0.2322215 0.3659722 1.0000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.1728  1.0000  0.6859  1.0000  1.0000

--- Summary for K = 4 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.4444  0.4794  0.4911  0.4921  0.5039  0.5444

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01389 0.03889 0.05129 0.07222 0.82222
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01444	0.04111	0.05293	0.07556	0.55222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.3517	0.4479	0.8978	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1161	0.7639	0.8044	0.8029	0.8439	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1744	0.1736	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.07278	0.49889	0.47684	0.83056	1.00000

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8744	0.9844	0.9878	0.9870	0.9911	0.9972

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01222	0.03111	0.05088	0.06389	0.87833

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01333	0.03333	0.05135	0.06764	0.59222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	3.860e-06	0.000e+00	1.789e-01

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	2.479e-07	0.000e+00	2.222e-03

Modality 4 :					
--------------	--	--	--	--	--

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	3.543e-07	0.000e+00	5.556e-04

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	1.058e-06	0.000e+00	1.278e-02

--- Summary for K = 6 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6400	0.7872	0.8217	0.8132	0.8467	0.8978

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.008889	0.025000	0.045040	0.052778	0.875556

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.010000	0.02667	0.04405	0.05611	0.49667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.04745	0.000000	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000000	0.00000000	0.00000000	0.0006463	0.00000000	0.6994444

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.00000	0.0050	0.00000	0.8889

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.03708	0.000000	1.000000

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8578	0.9811	0.9856	0.9845	0.9889	0.9961

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.007222 0.018333 0.037290 0.041667 0.883333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007778	0.020000	0.035654	0.044444	0.475000

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	5.510e-06	0.000e+00	3.128e-01

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00e+00	0.00e+00	0.00e+00	9.52e-05	0.00e+00	1.00e+00

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5144	0.5811	0.5972	0.5959	0.6122	0.6528

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.00500	0.02111	0.03250	0.04611	0.71444

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.005556 0.023333 0.034481 0.050000 0.468889
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.0000 0.0000 0.0179 0.0000 1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.3550 0.4883 0.5552 0.7450 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.01075 0.00000 0.44000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.04384 0.03389 1.00000

```

--- Summary for K = 9 ---

```

Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.8311 0.9139 0.9289 0.9255 0.9394 0.9633

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.003889 0.010000 0.018735 0.023194 0.795000
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1      1      1      1      1      1

```

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.004444 0.011111 0.019661 0.024444 0.357222
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00e+00 0.00e+00 0.00e+00 5.29e-05 0.00e+00 1.00e+00
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.02207 0.00000 1.00000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0      0      0      0      0      0
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.

```

```
0.000000 0.000000 0.000000 0.002031 0.000000 1.000000
', '\n`'))
```

```
## ‘‘‘
##
##
## --- Summary for K = 1 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.6850  0.7078  0.7144  0.7149  0.7222  0.7439
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.09278 0.13722 0.13204 0.16778 0.84556
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.09556 0.14000 0.13478 0.17111 0.76389
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.005000 0.008333 0.101966 0.031111 1.000000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0005556 0.0050000 0.0083333 0.0773990 0.0244444 1.0000000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.006111 0.015000 0.162394 0.118889 1.000000
##
## --- Summary for K = 2 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.4622  0.4967  0.5044  0.5040  0.5117  0.5367
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```

```

## 0.00000 0.04778 0.09722 0.09920 0.13611 0.91000
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.0500 0.1011 0.1020 0.1400 0.6922
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.000000 0.001667 0.058333 0.413768 1.000000 1.000000
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000000 0.0005556 0.0050000 0.2480509 0.3904167 1.0000000
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.1228 1.0000 0.6968 1.0000 1.0000
##
## --- Summary for K = 3 ---
## Z.ar:
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.4294 0.4550 0.4644 0.4642 0.4728 0.5067
##
## beta.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.02222 0.05944 0.06859 0.09833 0.89111
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:

```

```

## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.00000 0.02333 0.06222 0.07092 0.10222 0.60056
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.0000  0.4672  0.7611  0.6738  1.0000  1.0000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.7706  1.0000  1.0000  0.9998  1.0000  1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.0000000 0.0000000 0.0005556 0.2322215 0.3659722 1.0000000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.0000  0.1728  1.0000  0.6859  1.0000  1.0000
##
## --- Summary for K = 4 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.4444  0.4794  0.4911  0.4921  0.5039  0.5444
##
## beta.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.00000 0.01389 0.03889 0.05129 0.07222 0.82222
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
##      1      1      1      1      1      1
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
##      1      1      1      1      1      1
##
## gamma.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.00000 0.01444 0.04111 0.05293 0.07556 0.55222
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.0000  0.0000  0.3517  0.4479  0.8978  1.0000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.1161  0.7639  0.8044  0.8029  0.8439  1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.0000  0.0000  0.0000  0.1744  0.1736  1.0000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.00000 0.07278 0.49889 0.47684 0.83056 1.00000

```

```

##
## --- Summary for K = 5 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.8744 0.9844 0.9878 0.9870 0.9911 0.9972
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01222 0.03111 0.05088 0.06389 0.87833
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01333 0.03333 0.05135 0.06764 0.59222
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 3.860e-06 0.000e+00 1.789e-01
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 2.479e-07 0.000e+00 2.222e-03
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 3.543e-07 0.000e+00 5.556e-04
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 1.058e-06 0.000e+00 1.278e-02
##
## --- Summary for K = 6 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.6400 0.7872 0.8217 0.8132 0.8467 0.8978
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.008889 0.025000 0.045040 0.052778 0.875556
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

```

```

##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.00000 0.01000 0.02667 0.04405 0.05611 0.49667
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.00000 0.00000 0.00000 0.04745 0.00000 1.00000
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.0000000 0.0000000 0.0000000 0.0006463 0.0000000 0.6994444
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.0000 0.0000 0.0000 0.0050 0.0000 0.8889
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.00000 0.00000 0.00000 0.03708 0.00000 1.00000
##
## --- Summary for K = 7 ---
## Z.ar:
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.8578 0.9811 0.9856 0.9845 0.9889 0.9961
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.000000 0.007222 0.018333 0.037290 0.041667 0.883333
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.000000 0.007778 0.020000 0.035654 0.044444 0.475000
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.000e+00 0.000e+00 0.000e+00 5.510e-06 0.000e+00 3.128e-01

```

```

## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0      0      0      0      0      0
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0      0      0      0      0      0
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00e+00 0.00e+00 0.00e+00 9.52e-05 0.00e+00 1.00e+00
##
## --- Summary for K = 8 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.5144 0.5811 0.5972 0.5959 0.6122 0.6528
##
## beta.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00500 0.02111 0.03250 0.04611 0.71444
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.005556 0.023333 0.034481 0.050000 0.468889
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.0000 0.0179 0.0000 1.0000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.3550 0.4883 0.5552 0.7450 1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.01075 0.00000 0.44000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.04384 0.03389 1.00000
##
## --- Summary for K = 9 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.8311 0.9139 0.9289 0.9255 0.9394 0.9633
##

```

```

## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000000 0.003889 0.010000 0.018735 0.023194 0.795000
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000000 0.004444 0.011111 0.019661 0.024444 0.357222
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.00e+00 0.00e+00 0.00e+00 5.29e-05 0.00e+00 1.00e+00
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000000 0.000000 0.000000 0.02207 0.000000 1.000000
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      0      0      0      0      0      0
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000000 0.000000 0.000000 0.002031 0.000000 1.000000
##
## ‘‘‘

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1–9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.